

TaxBombAI: Open, Auditable Retirement Tax Exposure Projection

Jitendra Rout

Independent Researcher

[email/GitHub link]

September 2026

Abstract

Retirement tax planning tools that project Required Minimum Distributions (RMDs) and their associated tax exposure are widely offered by financial institutions, but are typically closed-source, making their underlying methodology impossible for an independent party to verify. Separately, a growing number of financial tools now layer large language model (LLM) explanations atop computed figures, raising a distinct trust question: does the generated narrative accurately reflect the numbers it purports to describe, or does it introduce unverified claims? This paper presents TaxBombAI, an open-source RMD and tax-exposure projection tool with an explicit architectural separation between a deterministic calculation engine and an LLM narration layer, together with three forms of independent validation: (1) cross-validation of the calculation engine against an independently re-implemented reference oracle across 500 randomized scenarios (10,318 individual year-level figures), finding exact agreement within rounding tolerance; (2) a parameter sweep characterizing how peak RMD magnitude and lifetime tax exposure scale with starting balance, age, and Roth conversion strategy; and (3) an automatic faithfulness verifier that checks whether AI-generated narrative text contains only numeric claims traceable to the underlying computed schedule, validated against constructed test narratives with known injected fabrications. We report this as evidence of the calculation engine’s correctness and of a workable verification approach for AI-narrated financial explanations, while being explicit that the faithfulness verifier has been validated against constructed test cases and not yet against live model output, which we identify as the clearest next step.

1 Introduction

Required Minimum Distribution (RMD) rules require holders of traditional tax-deferred retirement accounts to begin withdrawing, and paying tax on, a portion of their balance from a specified age onward. Because account balances compound for years before RMDs begin, the resulting tax liability can be substantial and is frequently larger than account holders anticipate — informally termed a “tax bomb” in financial planning discussion. Software tools that project this exposure are common, offered by financial institutions, advisors, and consumer finance sites; however, the substantial majority are closed-source, meaning a user or independent reviewer cannot inspect the underlying formula, verify it against the source IRS tables, or confirm the tool is free of implementation error.

A second, more recent trend compounds this opacity concern: financial tools increasingly layer an LLM-generated natural-language explanation on top of computed figures, intended to make the output more accessible to a non-technical user. This introduces a distinct failure mode beyond a possible calculation bug — the narrative layer could, in principle, state a figure that does not match what was actually computed, whether through model error or ungrounded elaboration. To our knowledge, open financial planning tools that layer AI narration rarely publish any mechanism for verifying that the generated narrative remains faithful to the underlying computation.

This paper makes three contributions. First, we describe TaxBombAI’s architecture, which enforces a strict separation between a deterministic calculation engine and an LLM narration layer instructed to narrate, not compute, so that the numeric output remains auditable independent of the AI layer. Second, we validate the calculation engine through independent cross-implementation: a from-scratch re-implementation of the RMD projection formula in a different language, checked against the production implementation across 500 randomized scenarios. Third, we introduce and validate an automatic faithfulness verifier — a non-AI, pattern-based checker that flags numeric claims in AI-generated narrative text unsupported by the underlying computed schedule — and report its validation against constructed test cases with deliberately injected fabrications, while being explicit that evaluation against live model output remains future work.

2 System Architecture

TaxBombAI consists of three components. The **calculation engine** (`src/calculator.js`) implements RMD projection using the IRS Uniform Lifetime Table (Publication 590-B, Table III), a configurable marginal tax bracket schedule, and a Roth conversion ladder comparison mode. It is a pure function of its inputs, with no external dependencies or network calls, making it independently testable and auditable. The **narration layer** (`src/explainer.js`) sends only the already-computed schedule to an LLM (Claude), with an explicit system instruction not to recompute or invent figures, and asks it to narrate the result in plain language. The **faithfulness verifier** (`src/faithfulnessVerifier.js`), introduced in this paper, provides a non-AI, deterministic check of the narration layer’s output: it extracts dollar-figure and age references from generated text via pattern matching and flags any that do not correspond, within a small tolerance, to a value actually present in the computed schedule.

This architectural separation is the paper’s central design claim: by construction, the numeric content of TaxBombAI’s output does not depend on the LLM at all, and the LLM’s role is strictly narrational. The faithfulness verifier provides an automatic, auditable check of that property rather than relying on it as an unverified assumption.

3 Validation Methodology

3.1 Independent Cross-Validation of the Calculation Engine

We wrote a second, independent implementation of the RMD projection formula in Python (`validation/reference_oracle.py`) by transcribing the IRS Uniform Lifetime Table divisors directly from Publication 590-B rather than copying the JavaScript implementation’s table. We then generated 500 randomized test scenarios, sampling starting age (45–70), traditional balance (\$50,000–\$5,000,000), annual growth rate (0–10%), RMD start age (73 or 75, per SECURE 2.0), projection horizon (10–30 years), and other taxable income (\$0–\$150,000) uniformly at random with a fixed seed for reproducibility. Each scenario was run through both the production JavaScript calculator and the independent Python oracle, and every individual year-level output (RMD amount, taxable income, estimated tax, ending balance) was compared between the two implementations.

3.2 Parameter Sweep

To characterize the calculation engine’s behavior across realistic input ranges, we swept starting age ($\{55, 60, 65, 70\}$) and starting traditional balance ($\{\$250\text{K}, \$500\text{K}, \$1\text{M}, \$2\text{M}, \$4\text{M}\}$) at fixed 6% growth, RMD start age 73, and \$40,000 other taxable income, recording the age and magnitude of peak annual RMD and cumulative lifetime tax for each combination. We separately swept the annual Roth conversion amount (\$0 to \$150,000/year) for a fixed profile (age 60, \$1.5M starting balance, conversion window ages 60–72) to characterize the lifetime-tax effect of the Roth conversion ladder comparison mode.

3.3 Faithfulness Verifier Validation

We constructed two test narratives structurally resembling AI-generated output: one in which every quoted dollar figure and age reference is drawn directly from a computed schedule (the "faithful" case), and one structurally identical narrative in which the peak RMD amount and peak RMD age were each replaced with a nearby but incorrect invented value (the "unfaithful" case, simulating a hallucinated or miscalculated figure). We ran the faithfulness verifier against both and checked whether it correctly classified each. We emphasize that neither narrative was generated by a live LLM call for this validation — the faithful narrative was authored to resemble expected model output, and the unfaithful narrative was constructed by deliberately injecting known-incorrect values into it. This validates the verifier's detection logic; it does not constitute an evaluation of the narration layer's actual behavior when connected to a live model, which we did not have API access to perform in this evaluation environment and identify as the clearest direction for follow-up validation.

4 Results

4.1 Cross-Validation

Across 500 randomized scenarios, comprising 10,318 individual year-level output rows, the JavaScript production implementation and the independently-written Python oracle agreed within a \$1 rounding tolerance on every compared value (RMD amount, taxable income, estimated tax, and ending balance) with zero mismatches. Table 1 summarizes the validation run.

Table 1: Cross-validation summary.

Randomized scenarios	500
Individual year-rows compared	10,318
Fields compared per row	4
Mismatches beyond \$1 tolerance	0

4.2 Parameter Sweep Findings

Figure 1 shows peak annual RMD as a function of starting balance across four starting ages. Two properties are apparent and directly follow from the model's structure: peak RMD magnitude scales linearly with starting balance at fixed age and growth rate (since RMD is computed as balance divided by an age-indexed divisor, and balance evolution is a linear recursion in the starting balance), while the *age* at which the peak occurs is invariant to starting balance, depending only on the growth rate, RMD start age, and the shape of the Uniform Lifetime Table divisor sequence. This is a useful sanity property for reviewing the model's internal consistency: any parameter sweep in which peak-RMD age varied with balance alone, holding other parameters fixed, would indicate an implementation error.

Figure 2 shows lifetime tax savings from the Roth conversion ladder comparison mode as a function of annual conversion amount, for a fixed profile (age 60, \$1.5M starting balance). Savings increase with conversion amount but with clearly diminishing marginal returns (\$96,238 saved at the first \$25,000/year increment vs. \$71,908 saved at the increment from \$100,000 to \$150,000/year), consistent with the expected mechanism: early conversions fill lower tax brackets efficiently, while larger conversions increasingly push into higher marginal rates, reducing the per-dollar benefit of further conversion.

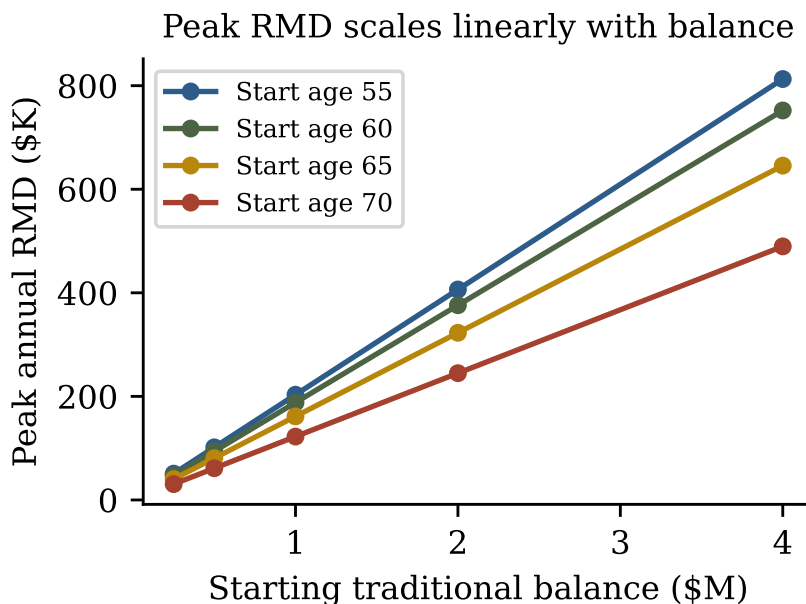


Figure 1: Peak annual RMD vs. starting balance, by starting age (6% growth, RMD start age 73).

4.3 Faithfulness Verifier

The faithfulness verifier correctly classified both constructed test cases: the faithful narrative (two dollar figures, two age references, all drawn from the computed schedule) was correctly flagged as faithful with zero unsupported values; the unfaithful narrative was correctly flagged as unfaithful, with the verifier identifying both injected fabrications (an incorrect dollar figure and an incorrect age reference) by name. We report this as validation of the verifier’s detection logic under a controlled, known-answer test, not as an evaluation of the narration layer’s real-world behavior.

5 Discussion

The cross-validation result (zero mismatches across 10,318 compared values between two independent implementations) provides meaningfully stronger evidence of the calculation engine’s correctness than testing a single implementation against its own internal logic would, since a systematic error shared by both a piece of code and its own unit tests would not be caught by that approach, whereas an independent re-implementation written from the source IRS tables directly provides an external check.

The parameter sweep’s structural finding — peak RMD age is invariant to starting balance while peak RMD magnitude scales linearly with it — is a useful property for anyone reviewing or extending this code-base: it provides a simple, checkable invariant that any correct implementation of this specific model should satisfy, independent of re-running the full cross-validation suite.

The faithfulness verifier represents, we believe, a reasonable minimum bar for AI-narrated financial tools: an automatic, non-AI check that narrated figures are traceable to the underlying computation, rather than trusting the narration layer’s fidelity as an unverified design assumption. We are explicit, however, that what we have validated here is the verifier’s ability to distinguish constructed faithful and unfaithful text — a necessary but not sufficient step toward validating the narration layer’s actual behavior in production, which requires live-model evaluation we did not have the infrastructure to perform in this work.

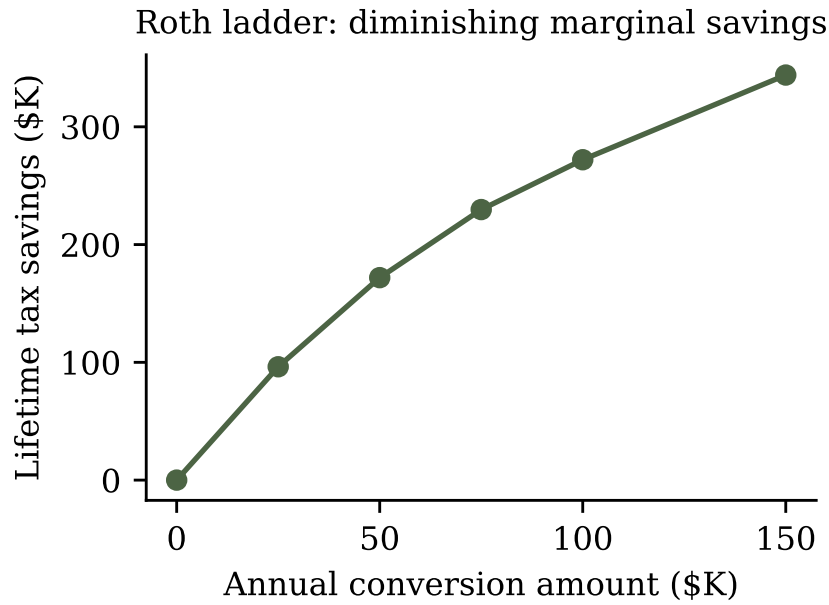


Figure 2: Lifetime tax savings vs. annual Roth conversion amount (age 60, \$1.5M starting balance, conversion window 60–72).

6 Limitations

- **Faithfulness verifier not yet evaluated against live model output.** As stated above, our validation used constructed test narratives, not real LLM-generated text. It is possible that real model output contains failure modes (e.g., correct figures presented with misleading qualitative framing, or figures that are technically present in the schedule but selectively misrepresented) that a purely numeric-matching verifier would not catch. Live-model evaluation, ideally across many real generations and multiple models, is the clearest next step.
- **Verifier scope.** The current verifier checks dollar figures and age references only; it does not verify percentage claims, qualitative characterizations, or causal claims about which strategy is better, which would require more sophisticated natural language understanding than pattern matching provides.
- **Tax model simplifications.** As documented in the project’s methodology notes, the bundled tax bracket table is illustrative and not auto-updated for the current tax year, and does not model state tax, NIIT, the Social Security taxability phase-in, or filing-status variations. The Roth conversion ladder comparison does not model the five-year rule, IRMAA thresholds, or sequence-of-returns risk.
- **Cross-validation covers the deterministic engine only.** The 500-scenario cross-validation validates the calculation engine’s correctness; it says nothing about the narration layer’s behavior, which is addressed separately (and only partially, per above) by the faithfulness verifier.

7 Conclusion

We presented TaxBombAI, an open-source RMD and tax-exposure projection tool built around an explicit separation between a deterministic, independently-validated calculation engine and an LLM narration layer,

together with an automatic faithfulness verifier for checking that narrated text remains traceable to the underlying computation. Cross-validation against an independently re-implemented oracle found zero mismatches across 10,318 compared values, and the faithfulness verifier correctly distinguished constructed faithful and unfaithful test narratives. We regard live-model evaluation of the faithfulness verifier against real narration-layer output as the most valuable direction for follow-up work, and release all code, validation scripts, and raw results openly to support that and other extensions.

Acknowledgments

The author acknowledges the use of Claude (Anthropic) in the development of this project, including drafting portions of the calculation engine, validation scripts, and this manuscript, which were reviewed, tested, and revised by the author. All reported validation results (cross-validation figures, parameter sweep output, and faithfulness verifier test outcomes) were produced by executing the accompanying code and are independently reproducible from the published source.

Availability

All code, validation scripts, and raw results described in this paper are published under the MIT license at <https://github.com/jitendrarout/taxbombai>.

References

- [1] Internal Revenue Service. *Publication 590-B: Distributions from Individual Retirement Arrangements (IRAs)*. <https://www.irs.gov/publications/p590b>
- [2] U.S. Congress. *SECURE 2.0 Act of 2022*. Required minimum distribution age provisions.